

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1.B

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Date:

1. TITLE

HTML5 Graphics and CSS3 Web Styling

2. AIM

To design and develop a styled webpage using HTML5 Canvas and CSS3, implementing graphics, navigation, content cards, and modern styling effects such as gradients, shadows, borders, rounded corners, and hover effects.

3. OBJECTIVE

- To understand HTML5 Canvas and basic graphic drawing.
 - To learn CSS3 selectors and styling properties.
 - To apply colors, fonts, borders, margins, padding, and backgrounds.
 - To implement gradients, shadows, rounded corners, and hover effects.
 - To create a visually appealing webpage using HTML5 and CSS3.
-

4. THEORY

HTML5 provides the Canvas element for drawing graphics directly on a webpage using JavaScript. Canvas can be used to draw shapes such as rectangles, circles, lines, and text.

CSS3 (Cascading Style Sheets) is used to control the appearance and layout of HTML elements. CSS selectors are used to target elements, while properties such as colors, fonts, margins, padding, borders, and backgrounds improve the webpage design.

CSS3 also provides advanced styling features such as gradients, border-radius, box-shadow, text-shadow, and hover effects. These features make webpages more attractive and interactive.

In this experiment, HTML5 Canvas is used to create basic graphics, while CSS3 is used to style the header, navigation bar, canvas section, three content cards, buttons, and footer.

Flow:

HTML5 Structure → Canvas Graphics → CSS Selectors → Basic Styling → Advanced CSS Effects → Hover Effects → Styled Webpage

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and link the external CSS file.
 2. Create the webpage structure with a header, navigation bar, main content area, and footer.
 3. Add a Canvas element and obtain its 2D drawing context using JavaScript.
 4. Draw a rectangle, circle, line, and text on the Canvas.
 5. Create three content cards with headings, descriptions, and buttons.
 6. Apply CSS selectors to style the webpage using colors, fonts, margins, padding, borders, and backgrounds.
 7. Apply gradients, rounded corners, box shadows, and text effects.
 8. Add hover effects to navigation links, cards, and buttons.
 9. Save both HTML and CSS files and open the HTML file in a browser.
 10. Verify the Canvas graphics, layout, styling, and hover effects.
-

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Main Container -->
  <div class="page">

    <!-- Header -->
    <header class="header">
      <div class="header-content">
        <h1>My Styled Webpage</h1>
```

```

        <p>HTML5 Canvas & CSS3 Web Styling</p>
    </div>
</header>

<!-- Navigation -->
<nav class="navbar">
    <div class="nav-container">
        <a href="#" class="active">Home</a>
        <a href="#">About</a>
        <a href="#">Gallery</a>
        <a href="#">Contact</a>
    </div>
</nav>

<!-- Main Content -->
<main class="main-content">

    <!-- Canvas Card -->
    <section class="canvas-card">
        <div class="section-title">
            <h2>HTML5 Canvas</h2>
        </div>
        <div class="canvas-box">
            <canvas id="myCanvas" width="700" height="250">
                Your browser does not support Canvas.
            </canvas>
        </div>
    </section>

    <!-- Cards Section -->
    <section class="cards-section">
        <div class="section-title">
            <h2>Content Cards</h2>
        </div>
        <div class="cards-container">

            <div class="card">
                <div class="card-image teal-grad-1"></div>
                <div class="card-content">
                    <h3>Card One</h3>
                    <p>This is the first content card with updated styling and layout.</p>
                    <button>Read More</button>
                </div>
            </div>

            <div class="card">
                <div class="card-image teal-grad-2"></div>
                <div class="card-content">
                    <h3>Card Two</h3>
                    <p>This is the second content card designed with smooth hover state
feedback.</p>
                    <button>Read More</button>
                </div>
            </div>
        </div>
    </section>

```

```
        <div class="card">
            <div class="card-image teal-grad-3"></div>
            <div class="card-content">
                <h3>Card Three</h3>
                <p>This is the third content card showcasing rounded corners and glass
accents.</p>
                <button>Read More</button>
            </div>
        </div>

    </div>
</section>

</main>

<!-- Footer -->
<footer class="footer">
    <p>© 2026 My Styled Webpage | HTML5 & CSS3</p>
</footer>

</div>

<script>
    const canvas = document.getElementById("myCanvas");
    const ctx = canvas.getContext("2d");

    // Rounded Green/Teal Rectangle
    ctx.fillStyle = "#0d9488";
    ctx.beginPath();
    ctx.roundRect(30, 30, 150, 80, 12);
    ctx.fill();

    // Circle with Cyan Glow
    ctx.beginPath();
    ctx.arc(280, 70, 40, 0, Math.PI * 2);
    ctx.fillStyle = "#14b8a6";
    ctx.fill();

    // Accent Line
    ctx.beginPath();
    ctx.moveTo(360, 30);
    ctx.lineTo(600, 120);
    ctx.strokeStyle = "#2dd4bf";
    ctx.lineWidth = 4;
    ctx.lineCap = "round";
    ctx.stroke();

    // Text
    ctx.font = "bold 26px Inter, sans-serif";
    ctx.fillStyle = "#f43f5e";
    ctx.fillText("Your Name", 220, 190);
</script>
```

```
</body>
```

```
</html>
```

Style.css

```
/*
=====
=====
1. RESET & BASE TEAL STYLES
=====
===== */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Inter', -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif;
  background-color: #041e23;
  /* Deep dark teal background */
  color: #e6f7f8;
  line-height: 1.6;
  min-height: 100vh;
}

.page {
  display: flex;
  flex-direction: column;
  min-height: 100vh;
}

/*
=====
=====
2. HEADER
=====
===== */
.header {
  background: radial-gradient(circle at top, #0d4a52 0%, #041e23 100%);
  padding: 64px 20px 48px;
  text-align: center;
  border-bottom: 1px solid rgba(45, 212, 191, 0.15);
}

.header-content h1 {
  font-size: 2.8rem;
  font-weight: 700;
```

```

letter-spacing: -0.01em;
background: linear-gradient(135deg, #2dd4bf, #a5f3fc);
-webkit-background-clip: text;
-webkit-text-fill-color: transparent;
margin-bottom: 8px;
}

.header-content p {
font-size: 1.1rem;
color: #99f6e4;
opacity: 0.85;
font-weight: 40;
font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida Grande', 'Lucida Sans
Unicode', Geneva, Verdana, sans-serif;
}

/*
=====
=====
3. NAVIGATION BAR
=====
===== */
.navbar {
position: sticky;
top: 0;
z-index: 100;
background: rgba(4, 30, 35, 0.85);
backdrop-filter: blur(14px);
-webkit-backdrop-filter: blur(14px);
border-bottom: 1px solid rgba(45, 212, 191, 0.12);
}

.nav-container {
max-width: 1100px;
margin: 0 auto;
display: flex;
justify-content: center;
gap: 8px;
padding: 12px 20px;
}

.nav-container a {
text-decoration: none;
color: #ccfbf1;
font-weight: 500;
font-size: 0.95rem;
padding: 8px 20px;
border-radius: 99px;
transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
}

.nav-container a:hover,
.nav-container a.active {

```

```

    color: #041e23;
    background: #2dd4bf;
    box-shadow: 0 0 15px rgba(45, 212, 191, 0.35);
}

/*
=====
=====
    4. MAIN CONTENT
=====
===== */
.main-content {
    max-width: 1100px;
    width: 100%;
    margin: 40px auto;
    padding: 0 20px;
    display: flex;
    flex-direction: column;
    gap: 48px;
    flex: 1;
}

.section-title h2 {
    font-size: 1.5rem;
    font-weight: 700;
    color: #f0fdfa;
    margin-bottom: 20px;
    position: relative;
    display: inline-block;
}

.section-title h2::after {
    content: "";
    position: absolute;
    bottom: -6px;
    left: 0;
    width: 36px;
    height: 3px;
    background: #14b8a6;
    border-radius: 2px;
}

/*
=====
=====
    5. CANVAS SECTION
=====
===== */
.canvas-card {
    background: #0b2b30;
    border-radius: 20px;
    padding: 28px;

```

```

border: 1px solid rgba(45, 212, 191, 0.12);
box-shadow: 0 12px 32px -10px rgba(0, 0, 0, 0.6);
}

.canvas-box {
  display: flex;
  justify-content: center;
  align-items: center;
  background: #031518;
  border-radius: 14px;
  padding: 20px;
  border: 1px dashed rgba(45, 212, 191, 0.25);
  overflow-x: auto;
}

canvas {
  max-width: 100%;
  height: auto;
  display: block;
  background-color: #0d1f23;
  border-radius: 10px;
  border: 1px solid rgba(255, 255, 255, 0.05);
}

/*
=====
=====
6. CARDS SECTION
=====
===== */
.cards-container {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
  gap: 24px;
}

.card {
  background: #0b2b30;
  border-radius: 20px;
  overflow: hidden;
  border: 1px solid rgba(45, 212, 191, 0.12);
  transition: transform 0.3s cubic-bezier(0.4, 0, 0.2, 1),
    box-shadow 0.3s cubic-bezier(0.4, 0, 0.2, 1),
    border-color 0.3s ease;
  display: flex;
  flex-direction: column;
}

.card:hover {
  transform: translateY(-8px);
  box-shadow: 0 20px 35px -10px rgba(0, 0, 0, 0.5), 0 0 20px rgba(20, 184, 166, 0.15);
  border-color: rgba(45, 212, 191, 0.4);
}

```

```
.card-image {
  height: 140px;
  border-bottom: 1px solid rgba(45, 212, 191, 0.1);
}

.teal-grad-1 {
  background: linear-gradient(135deg, #115e59, #0d9488);
}

.teal-grad-2 {
  background: linear-gradient(135deg, #0f766e, #14b8a6);
}

.teal-grad-3 {
  background: linear-gradient(135deg, #134e4a, #0f766e);
}

.card-content {
  padding: 24px;
  display: flex;
  flex-direction: column;
  gap: 12px;
  flex: 1;
}

.card-content h3 {
  font-size: 1.25rem;
  font-weight: 600;
  color: #f0fdfa;
}

.card-content p {
  color: #99f6e4;
  opacity: 0.75;
  font-size: 0.95rem;
  flex: 1;
}

.card-content button {
  background: #14b8a6;
  color: #041e23;
  border: none;
  padding: 10px 22px;
  border-radius: 10px;
  font-weight: 600;
  font-size: 0.9rem;
  cursor: pointer;
  transition: all 0.2s ease;
  align-self: flex-start;
}

.card-content button:hover {
  background: #2dd4bf;
}
```

```

    box-shadow: 0 0 12px rgba(45, 212, 191, 0.4);
}

.card-content button:active {
    transform: scale(0.97);
}

/*
=====
=====
7. FOOTER
=====
===== */
.footer {
    text-align: center;
    padding: 32px 20px;
    border-top: 1px solid rgba(45, 212, 191, 0.1);
    color: #5eead4;
    opacity: 0.6;
    font-size: 0.9rem;
    margin-top: auto;
}

```

7. CODE EXPLANATION

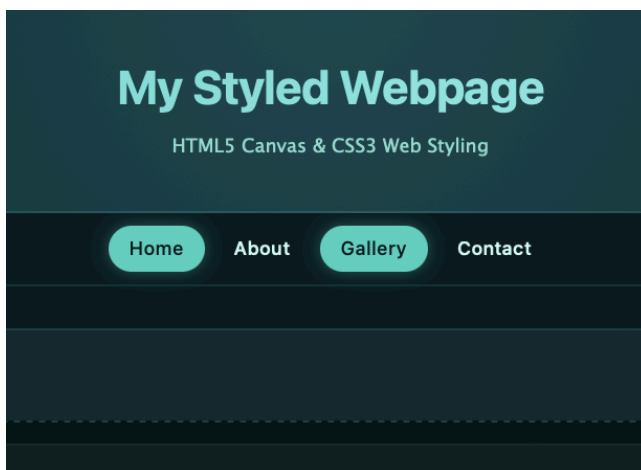
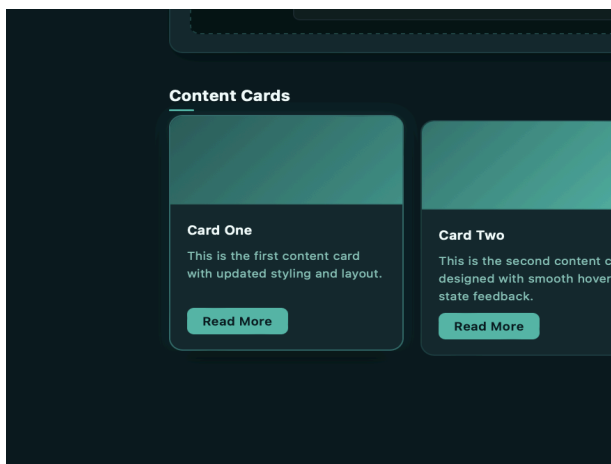
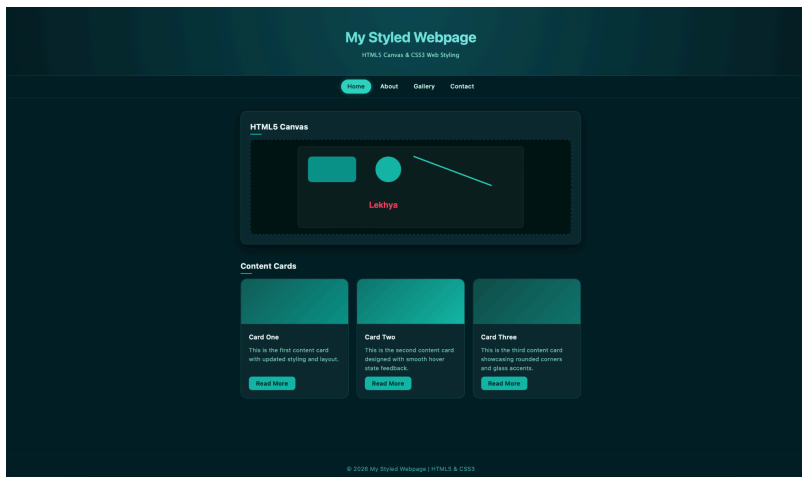
HTML / CSS / JS Code	Purpose
<code><!DOCTYPE html></code>	Declares the document as HTML5.
<code><link rel="stylesheet" href="style.css"></code>	Connects the external CSS file.
<code><header></code>	Creates the webpage header.
<code><nav></code>	Creates the navigation bar.
<code><canvas></code>	Provides an area for drawing graphics.
<code>getContext("2d")</code>	Gets the 2D drawing context of the Canvas.
<code>ctx.roundRect()</code>	Draws the rectangle on the Canvas.
<code>ctx.arc()</code>	Draws the circle.

<code>ctx.moveTo()</code> and <code>ctx.lineTo()</code>	Draws the line.
<code>ctx.fillText()</code>	Displays text on the Canvas.
<code>.card</code>	Styles the three content cards.
<code>linear-gradient()</code> / <code>radial-gradient()</code>	Creates gradient backgrounds.
<code>border-radius</code>	Creates rounded corners.
<code>box-shadow</code> / <code>text-shadow</code>	Adds shadow effects.
<code>:hover</code>	Applies effects when the user moves the mouse over elements.
<code><footer></code>	Creates the footer of the webpage.

8. TEST CASES AND EXECUTION DATA

Test Case	Test Description	Expected Result	Status
TC01	Open the HTML file	Styled webpage loads correctly	Pass
TC02	Check Canvas graphics	Rectangle, circle, line, and text are displayed	Pass
TC03	Check cards and CSS styling	Cards, gradients, shadows, and rounded corners appear correctly	Pass
TC04	Check hover effects	Navigation links, cards, and buttons respond on hover	Pass

9. OUTPUT



10. RESULT

The “**My Styled Webpage**” was successfully developed using HTML5 Canvas and CSS3. Basic graphics and modern styling effects such as gradients, shadows, rounded corners, and hover effects were implemented successfully.